

Question 2

→ Algorithm

- current floor $\leftarrow 0$
- Take input for n floors
- For every n floor if current $> n \rightarrow$ moving down, current $< n \rightarrow$ moving up, current $= n \rightarrow$ opening doors
- current $\leftarrow n$

→ PAC & IPO chart

Input	Process	output
N number of request	• Loop N times, current $\leftarrow 0$ • current floor $\leftarrow N[1]$ • If current $>$ floor $\rightarrow$ moving down • current $<$ floor $\rightarrow$ moving up • current = floor $\rightarrow$ opening door • current $\leftarrow$ floor	elevator positions/status
Floor		

Date: _____

Pseudocode:

~~000~~ N[] : ARRAY [0: n] OF INTEGER

current ← 0

FOR count ← 0 TO n

request ← N[count]

IF request > current THEN

OUTPUT "Moving Up"

Else IF request < current THEN

OUTPUT "Moving down"

Else

OUTPUT "Opening doors"

current ← request

ENDFOR

list [4, 3, 2, 0, 5]

Flowchart

START

INPUT

CUR

IF

request

0

0

Date:

Date:

Flowchart

